



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin
Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

Integrated Computer Science
Computer Science and Business
MSISS

Semester 2

Information Management II

27th April 2023

RDS-SIM COURT

14.00-16.00

Prof Yvette Graham; Prof Vincent Wade

Instructions to Candidates:

Attempt **three** questions in total. **Question 1 is mandatory.** Answer **any two** questions from **Question 2, Question 3 and Question 4**

All questions carry equal marks. Each question is scored out of a total of 25 marks.

Do not begin the exam until instructed to do so by an invigilator.

Materials permitted: None

Q. 1

A cinema chain is made up of a large number of movie theatres spread out across the country in different locations. Each theatre has its own schedule of movie screenings that is updated on a weekly basis. The database needs to store any information relevant to the movie screenings, such as start time, date, running time, movie title, movie director, and so on, as well as the information needed for the theatres, such as address, screens, amongst others.

- i) Develop an ER model for the above problem, stating clearly any assumptions you make.
[8 marks]
- ii) Using the appropriate mapping techniques, map the ER model to a Relational Model showing the functional dependency between objects represented there, the primary and foreign keys.
[5 marks]
- iii) Give the SQL commands for the following:
 - a. Insert a new movie "Titanic 2" into the database with MOVIE_ID=1012 including appropriate values for each of the attributes.
[3 marks]
 - b. Find the name of all of the movies with a running time of less than 1 hour 30 minutes with a release date in 2019.
[2 marks]
 - c. Find all of the movie screenings on today's date showing before 12 o'clock.
[3 marks]
 - d. Find all of the movie screenings on today's date *finishing* before 12 o'clock.
[4 marks]

Q 2.

- i) Describe three types of integrity constraints that can be applied to relational databases. [6 marks]
- ii) Describe how referential integrity can be specified in SQL giving an example of your choice with a clear explanation of why referential integrity constraints are appropriate. [6 marks]
- iii) Consider the below relations in a database that keeps track of stock levels in a music and movie store. Write an SQL statement that creates an assertion called `maximum_inventory` that prevents insertions into either table violating the following condition: the sum of the `sale_price` of all movies and music combined must not exceed 1,000,000 euro. [9 marks]

MOVIE

<u>ID</u>	Title	Director	Genre	Cost_Price	Sale_Price	...
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MUSIC

<u>ID</u>	Title	Artist	Genre	Cost_Price	Sale_Price	...
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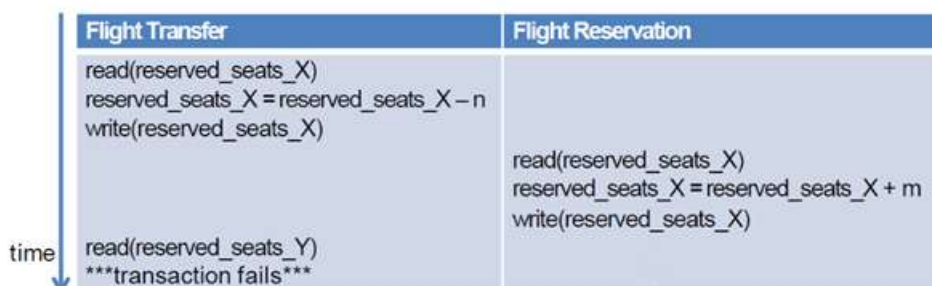
- iv) Explain any differences you are aware of between assertions and triggers in relational databases. [4 marks]

Q 3.

- i) Explain what is a transaction and how transactions can fail. [8 marks]
- ii) What are the ACID properties? Describe in detail. [8 marks]

- iii) In the below example, a Flight transfer transaction (between flights X and Y) and a Flight reservation (flight X only) transaction are being carried out simultaneously on the same database. The flight transfer transaction fails after reading data item reserved_seats_Y. Explain the type of error that will occur in this case.

[9 marks]



Q 4.

A prominent telecommunications company, Telemate, based in Ireland, is eager to get its latest deal out to its valued customers. An email list is set up including the email addresses of all existing customers. The offer that Telemate is putting together will be highly cost-effective for customers and the marketing manager, Helen, would like this information to reach as many customers as possible, as quickly as possible. An email is composed that highlights the most important parts of the offer, including a deal that will allow contract customers to own the latest iPhone at 50% of the usual contract price. Helen instructs Tom to send out the email to all existing customers on October 6th 2020. One customer, John, receives the email along with several other emails that John wastes time deleting, he is not interested in buying an iPhone. John would like to avoid receiving marketing emails altogether and never agrees to receiving these kinds of emails from companies.

- i) Describe in detail what John's rights are in terms of GDPR and what action John should initially take.

[8 marks]

- ii) Who in Telemate is responsible for the potential violation of GDPR?

[8 marks]

- iii) Despite John carrying out the action you described in (i), John was not satisfied with the result and in fact received a second similar email about a different offer 6 weeks later. What action can John now take and what are the possible repercussions for Telemate?

[9 marks]